

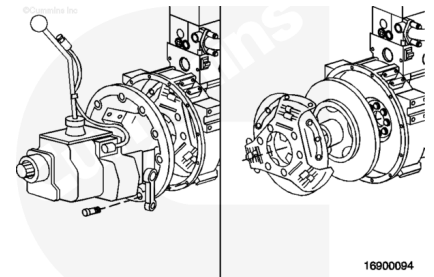
Trainings ▾

Remove

Remove the vehicle driveline and transmission. Refer to the manufacturer's instructions.

Remove the clutch discs and the pressure plate. Refer to the manufacturer's instructions.

Note : Use the barring tool, Part Number 3824591, to hold the flywheel to prevent rotation.



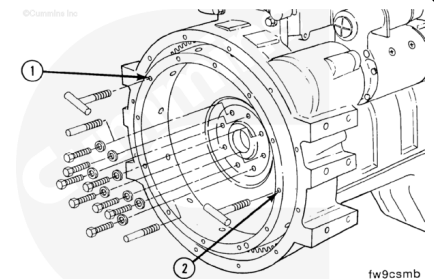
Remove two capscrews 180 degrees apart.

Install two M12 x 1.25 x 90-mm guide pins.

Note : If a clutch is used in the equipment, the threads in the clutch pressure plate mounting capscrew holes can be metric or standard. Be sure to use the correct capscrews.

Determine the capscrew thread design and size, and install two t-handles in the flywheel at points (1 and 2).

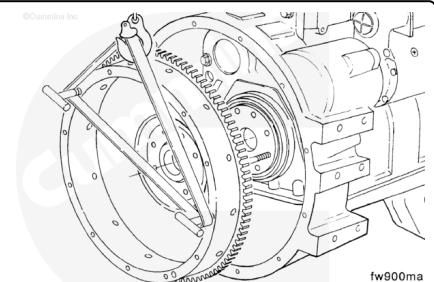
Remove the remaining six flywheel mounting capscrews.



⚠ WARNING ⚠

The component weighs 23 kg [51 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

Remove the flywheel from the guide pins.



Clean

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

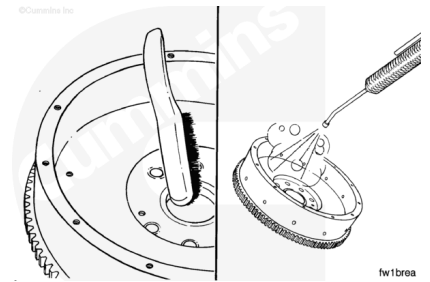
⚠ WARNING ⚠

Compressed air used for cleaning should not exceed 207 kPa [30 psi]. Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.

Use a wire brush to clean the crankshaft pilot bore.

Use steam or solvent to clean the flywheel.

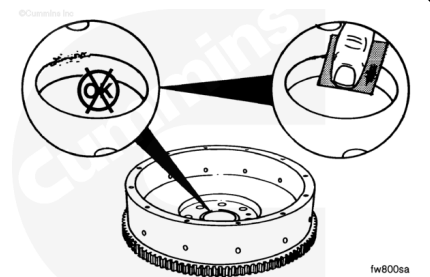
Dry with compressed air.



Inspect for Reuse

Inspect for nicks or burrs.

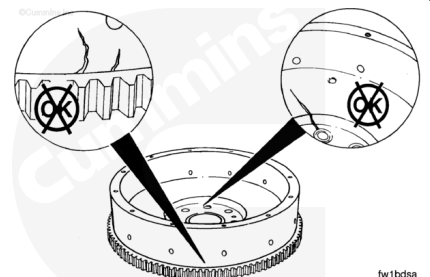
Use Scotch-Brite™ 7448, or equivalent, to remove small nicks and burrs.



⚠ WARNING ⚠

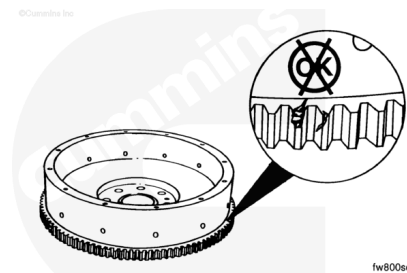
Do not use a cracked or resurfaced flywheel. These can break, causing serious personal injury or property damage.

Use the crack detection kit, Part Number 3375432, to check for cracks in the flywheel. Follow the instructions provided with the kit.



Inspect the flywheel ring gear teeth for cracks and chips.

Note : If the ring gear teeth are cracked or broken, the ring gear must be replaced. Refer to Procedure 016-008 (/qs3/pubsys2/xml/en/procedures/100/100-016-008.html).



Measure

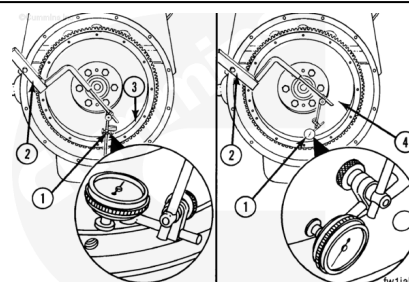
Flywheel Bore Runout

Use the dial indicator gauge (1), Part Number 3376050, or its equivalent, and dial gauge attachment (2), Part Number ST-1325, to inspect the flywheel bore (3) and the surface (4) runout.

Install the attachment to the flywheel housing.

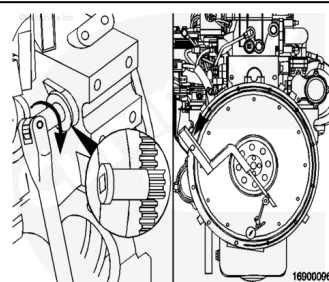
Install the gauge on the attachment.

Install the contact tip of the indicator against the inside diameter of the flywheel bore, and set the dial indicator at zero.



Use the barring tool, Part Number 3824591, to rotate the crankshaft one complete revolution.

Note : The total indicator reading (TIR) must not exceed 0.127 mm [0.0050 in].

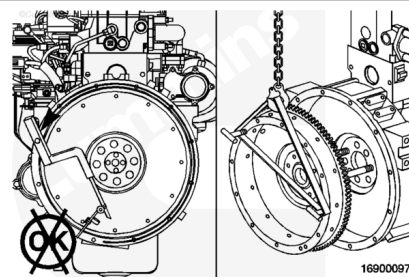


⚠ WARNING ⚠

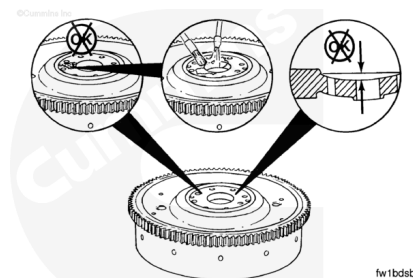
The component weighs 23 kg [51 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

Note : If the total indicator reading (TIR) is greater than the specification, do the following:

- Remove the flywheel.

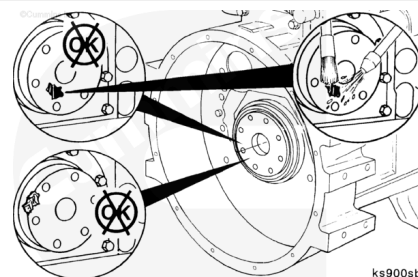


- Inspect the flywheel mounting surface for dirt or damage.



fw1bdsb

- Inspect the crankshaft for dirt or damage.

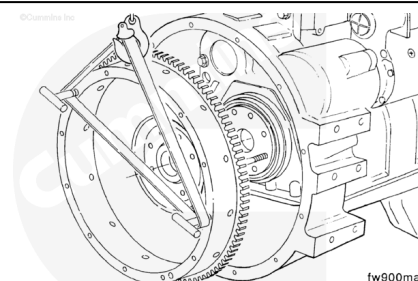


ks900sb

⚠ WARNING ⚠

The component weighs 23 kg [51 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

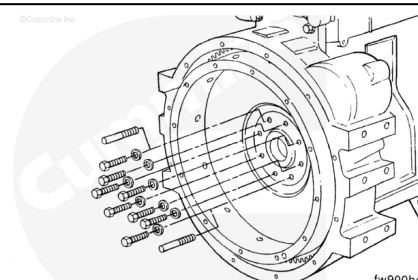
- Install the flywheel.
- Inspect the bore runout again.



fw900ma

- Replace the flywheel if the runout does **not** meet specifications.

mm		in
0.127	MAX	0.005

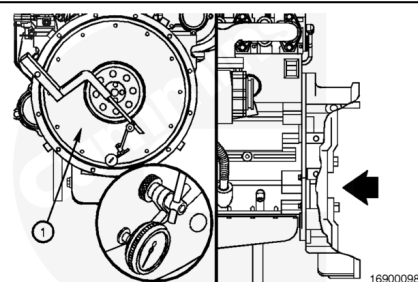


fw900he

Flywheel Face Runout

Install the contact tip of the indicator against the flywheel face, as close to the outside diameter as possible, to inspect the face (1) runout.

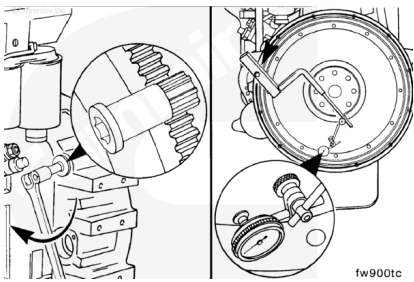
Note : Push the flywheel forward to remove the crankshaft end clearance. Adjust the dial on the indicator until the needle points to zero.



16900098

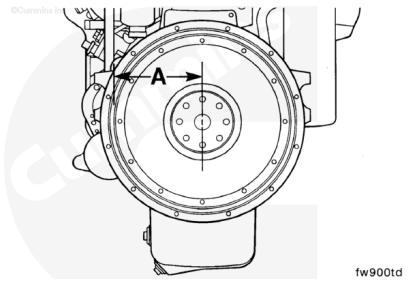
Use the barring tool, Part Number 3824591, to rotate the crankshaft one complete revolution. Measure the flywheel runout at four equal points on the flywheel.

Note : The flywheel must be pushed toward the front of the engine to remove the crankshaft end clearance each time a point is measured.

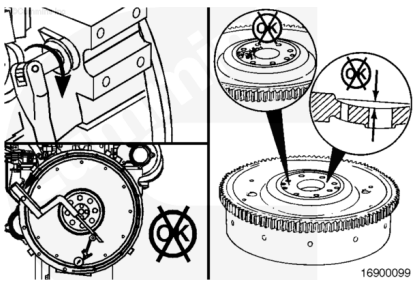


The total indicator reading (TIR) **must not** exceed the following specifications:

Flywheel Radius (A)		Maximum of Flywheel Face	
mm	in	mm	in
203	8	0.203	0.008
254	10	0.254	0.010
305	12	0.305	0.012
356	14	0.356	0.014
406	16	0.406	0.016

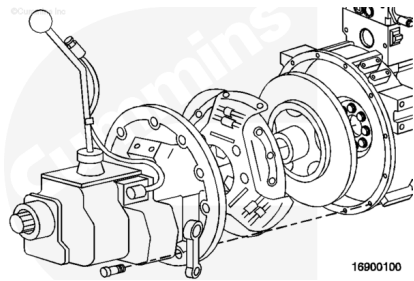


If the flywheel face runout is **not** within specifications, remove the flywheel. Check for nicks, burrs, or foreign material between the flywheel mounting surface and the crankshaft flange.



Install the clutch discs, pressure plate, transmission, and driveline (if equipped) in reverse order of removal. Refer to the manufacturer's instructions.

Note : Align the universal joints on each end of the driveshaft to prevent vibration.

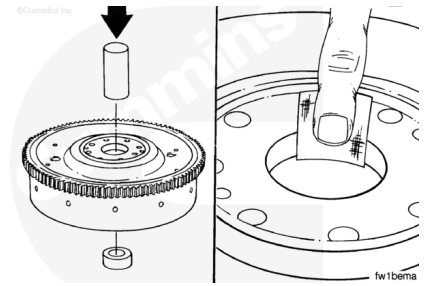


Install

Note : Use a new pilot bearing when installing a new or rebuilt clutch.

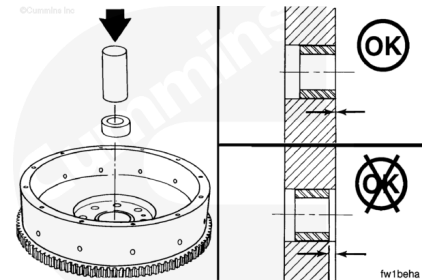
Use a mandrel and hammer to remove the pilot bearing.

Use a Scotch-Brite™ 7448, or equivalent, to clean the pilot bore.



Use a mandrel and hammer to install the pilot bearing.

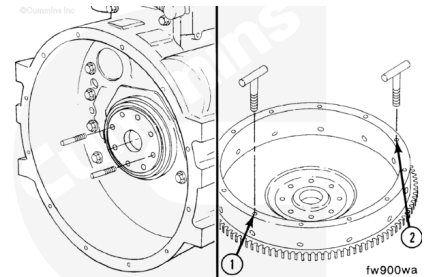
Note : The pilot bearing must be installed evenly with the pilot bore surface.



Install two M12 x 1.25 x 90-mm guide pins into the crankshaft flange 180 degrees apart.

Note : If a clutch is used in the equipment, the threads in the clutch pressure plate mounting capscrew holes can be metric or standard. Be sure to use the correct capscrews.

Determine the capscrew thread design and size, and install two t-handles into the flywheel (at points 1 and 2).

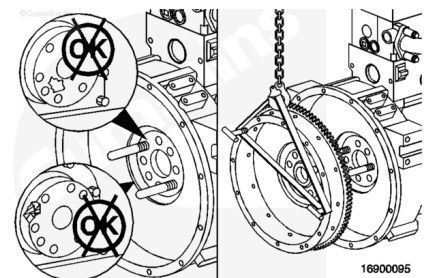


⚠ WARNING ⚠

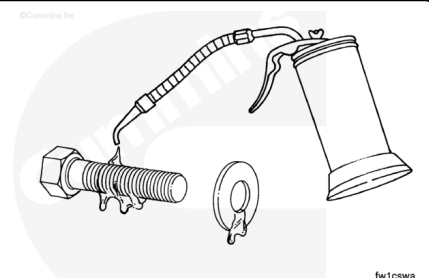
The component weighs 23 kg [51 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift the component.

Inspect the rear face of crankshaft and flywheel mounting flange for cleanliness and raised nicks or burrs.

Install the flywheel on the guide pins.



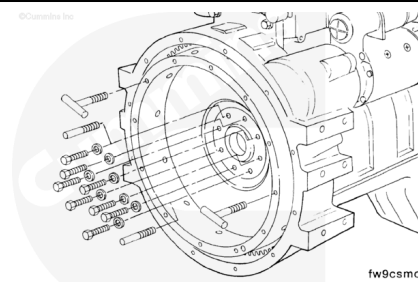
Lubricate the threads of the capscrews and the surface of the washers with clean lubricating engine oil.



Install the six capscrews.

Remove the t-handles and guide pins.

Install the remaining capscrews into the holes from which the guide pins were removed.



Barring Tool, Part Number 3824591

Hold the crankshaft when tightening the flywheel capscrews.

Tighten the capscrews in a star pattern.

Torque Value: 137 n•m [101 ft-lb]

Refer to the equipment manufacturer's procedures to install the transmission.

